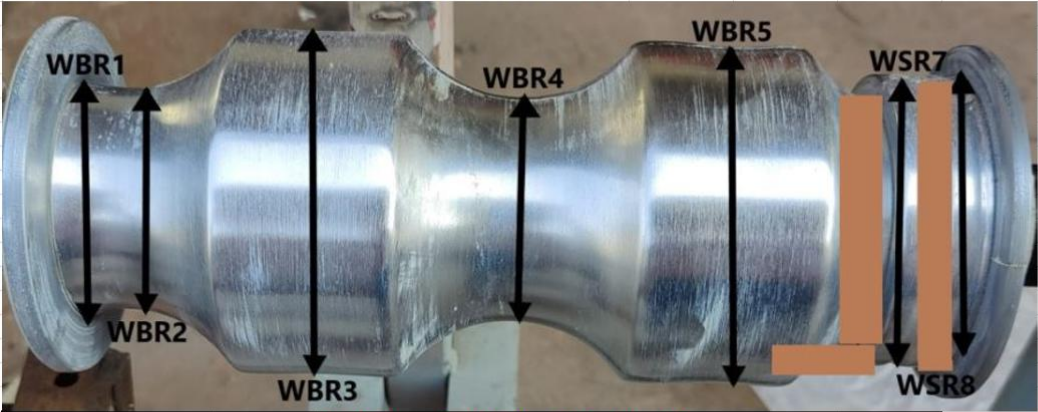


1. ROLLERS: The Foundation of the Process

I consider the rollers to be the starting point, as their condition directly impacts the performance of the other components.

DOUBLE ROMAN ROLLERS (AFRISTAR)



Big Roller + Small Roller 299 + 31 = 330 mm

BIG / SMALL DOUBLE ROMAN ROLLERS MEASURES

| TC BIG ROLLER - Nº |        |      |        |      | TC SMALL ROLLER - Nº |     |        |      |          |       |
|--------------------|--------|------|--------|------|----------------------|-----|--------|------|----------|-------|
|                    | WBR1   | WBR2 | WBR3   | WBR4 | WBR5                 |     | WSR7   | WSR8 |          |       |
| NEW                | 114,63 | 84   | 134    | 84   | 134                  | NEW | 126    | 122  | CHECK    |       |
| WEAR               |        |      | 128,00 |      | 128,00               |     | 121,00 | 117  | POSSIBLE | CHECK |

Big Roller (WBR3/5): A maximum wear reference we typically use is around 6 mm on its flat areas.

Small Roller (WSR7/8): Similarly, the maximum wear reference is typically 5 mm on its flat areas.

Impact and Relationship Between Components:

If the ranges indicated in the table (128/128/121/117 mm) are not respected, two main issues may arise:

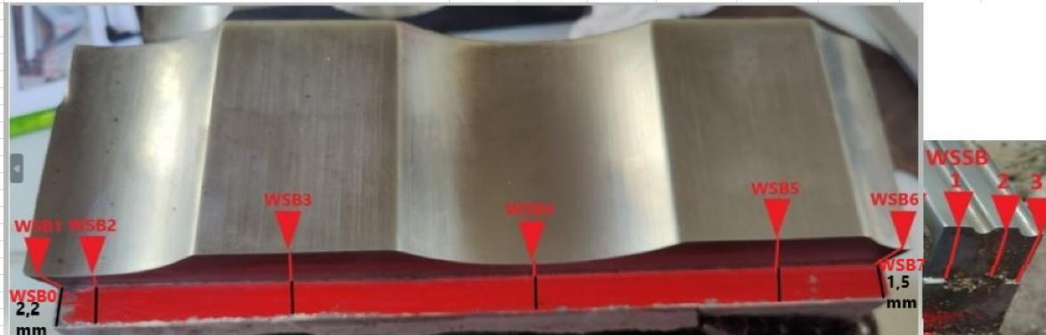
- Insufficient Feeding: The material does not reach the slippers with adequate consistency.
- Accelerated Wear of the Slippers: Without proper preforming, the slippers are forced to work harder in the final shaping, significantly reducing their lifespan.



## 2. SLIPPERS: Defining the Final Product

For slippers, the criteria must be more rigorous, as they are directly responsible for the final upper profile of the tile.

### DOUBLE ROMAN SLIPPERS (AFRISTAR)



| SLIPPER BIG- Nº (299 mm) |  |  |  |  |  |  | SLIPPER SMALL- Nº (31 mm) |  |  |  |
|--------------------------|--|--|--|--|--|--|---------------------------|--|--|--|
|--------------------------|--|--|--|--|--|--|---------------------------|--|--|--|

### BIG / SMALL DOUBLE ROMAN ROLL WEAR MEASURES

|            | WSB0  | WSB1 | WSB2 | WSB3 | WSB4 | WSB5 | WSB6 | WSB7  |            | WBSSB1 | WBSSB2 | WBSSB3 |       |
|------------|-------|------|------|------|------|------|------|-------|------------|--------|--------|--------|-------|
| NEW        | 34,51 | 34   | 20   | 45   | 20   | 45   | 37,5 | 37,56 | NEW        | 22,5   | 22,5   | 17,5   | CHECK |
| IN SUPPORT |       | 29   | 15   | 40   | 15   | 40   | 32,5 |       | IN SUPPORT | 22,5   | 23,5   | 17,5   | CHECK |
| RANGE      |       |      |      | 38,5 |      | 38,5 |      |       |            | 21,00  | 22,00  | 16,00  | CHECK |

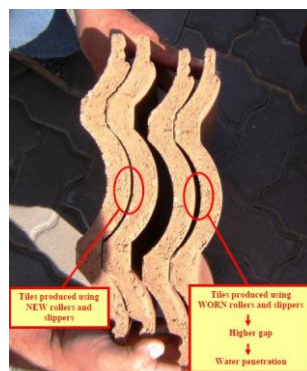
**Big Slipper (WSB3/5):** The maximum wear reference typically considered is  $\approx 1.5$  mm in the deformation of the tile's upper profile.

**Small Slipper (WSSB1/2):** Similarly,  $\approx 1.5$  mm (in practice, this is even more critical for the reasons explained below).

### Consequences of Wear and Replacement Criteria:

When the **Big Slipper (WSB3/5)** progressively alters the upper profile of the tile, it creates a fitting issue, as illustrated in the following photo.

Although this corresponds to a profile different from the Double Roman, it serves as a valid representation of the effect:





**The lower profile (defined by the mould) is fixed and invariable.**

**The upper profile, shaped by the worn Big Slipper, increases the tile thickness and creates gaps in the overlap once installed on the roof.**

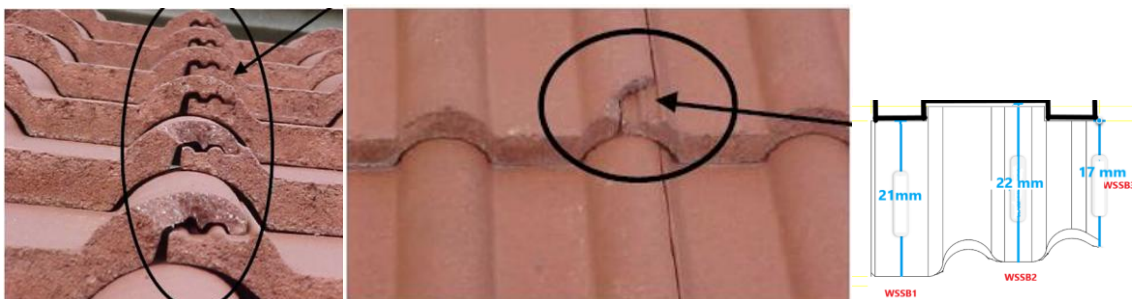
**This misalignment typically has two critical consequences:**

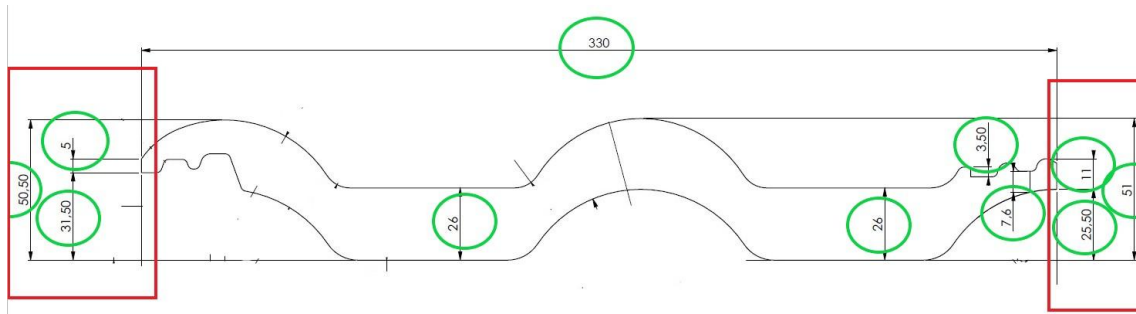
- Loss of Watertightness: The gaps allow water ingress, compromising the primary function of the roof.
- Replacement Criterion: The decision to replace should not be based solely on a measurement of Slipper wear but on the loss of the optimal profile.

**To determine the acceptable wear range and the maximum tolerable deformation, the following checks could be carried out:**

- Inspection and Monitoring of Slipper wear, comparing it with the reference ranges **(38.5/38.5/21/22/16 mm)**.
- Assembly of Test Roof Sections with the produced tiles to visually and functionally assess the extent of admissible deformation and establish the correct reference range for each case.
- As a complementary measure, it may be appropriate to use a control gauge to help define and quickly verify the maximum allowable surface deformation.

For its part, the **Small Slipper (WSSB1/2)** is equally critical. Its wear, by also increasing the tile thickness, reduces the clearance in the fitting. In my experience, this can lead to breakage issues when walking on the roof.





### **Key Relationship Between Components:**

Based on accumulated experience, the following wear relationship in component lifespan is typically observed:

**For every 1 Big Roller worn, 2 Small Rollers are usually worn.**

**For every 1 Big Roller worn, 2 Big Slippers are usually worn.**

**For every 1 Small Roller worn, 2 Small Slippers are usually worn.**

**In other words, as a general rule, the lifespan of a set typically corresponds to: 1 Big Roller, 2 Small Rollers, 2 Big Slippers, and 4 Small Slippers.**

**This chain relationship underscores the importance of coordinated preventive maintenance: uncontrolled wear in the rollers has a multiplier effect and accelerates the wear of the Slippers.**

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### **Conclusion**

**Keeping both Slippers and Slippers within verified wear ranges—always adapted to the specific raw materials of each factory—is not just about extending the lifespan of the parts.**

**I believe it is the most robust strategy for ensuring the final product quality, preventing water ingress, and avoiding future breakages during installation and roof maintenance.**